

ControlPlane.ai

Zero-Trust Guardrail Layer for Enterprise AI | Accenture Innovation Challenge 2026

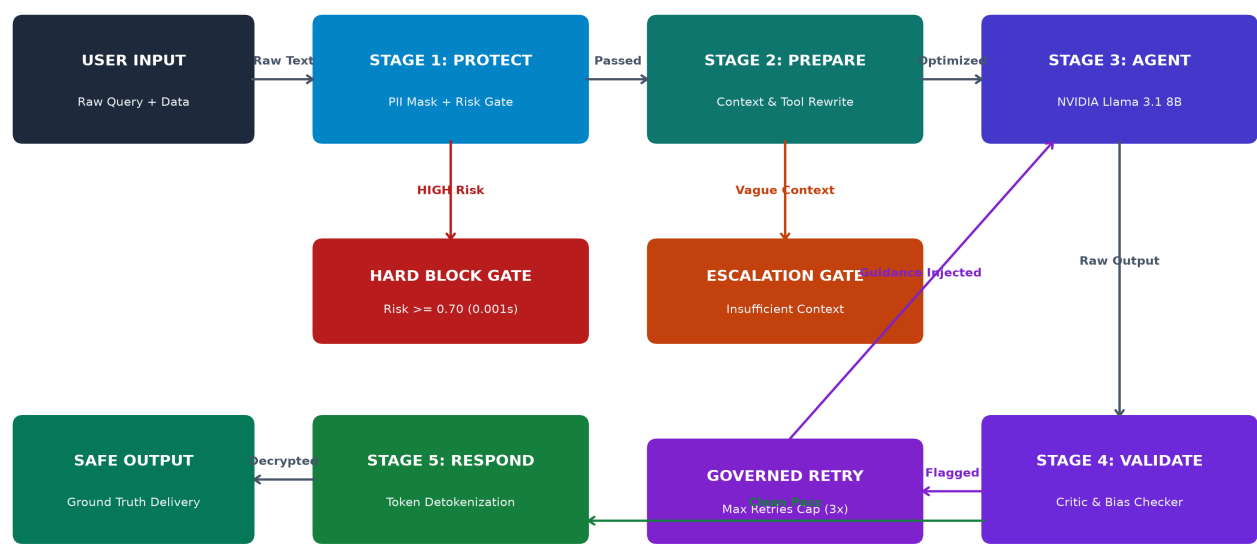
Author / Team: Grown Wings	Target Track: Enterprise AI Governance & Safety
Primary Model: NVIDIA Llama 3.1 8B (NIM)	Verified Net Savings: ~52.9% vs Frontier Models

1. Problem Statement & Enterprise Challenges

As organizations transition from passive copilots to autonomous multi-agent systems, they encounter an enterprise AI trilemma: **(1) Privacy & Secret Leaks** (accidental exposure of PII, API tokens, and credentials to third-party model providers), **(2) Factual Hallucinations & Policy Bias** (unverified model assertions and demographic stereotyping causing financial/reputational damage), and **(3) Runaway Compute Costs** (over-reliance on costly frontier LLMs for routine workflows without query optimization).

2. Solution Architecture: Observe, Evaluate, Act

ControlPlane.ai is a model-agnostic middleware sitting directly between end users and enterprise agents. It enforces a four-stage pipeline ensuring zero plaintext exposure, deterministic risk gating, and governed validation.



3. Stage-by-Stage Operational Breakdown

Stage 1: Protect (PII / Secret Masking + Risk Gate)

- **Tokenization Engine:** Replaces emails, phones, SSNs, credit cards, and high-entropy secrets (AWS keys, JWTs) with deterministic tokens (<PII_EMAIL_1>, <SECRET_API_KEY_1>).
- **Authoritative Risk Gate:** Evaluates prompt injection, jailbreaks, and wire fraud via deterministic rules. Scores >= 0.70 trigger a **HARD BLOCK** in <1ms (\$0.00 model compute).

Stage 2: Prepare (Context Check + Tool-Aware Rewrite)

- **Input 0 Tool Discovery:** Dynamically discovers available agent capabilities.
- **Context Sufficiency Check:** Identifies vague pronouns (it, that, them) and missing IDs, escalating for human clarification before invoking models.
- **Tool-Aware Query Rewrite:** Strips conversational fluff and injects required tool signatures ([TOOLS: ...]), compressing input tokens.

Stage 3: Enterprise Agent Inference (NVIDIA NIM Llama 3.1 8B)

- **Model-Agnostic Invocation:** Executes inference via OpenAI-compatible endpoints using Small Language Model (SLM) economics (~\$0.18/1M tokens).
- **Zero-Trust Safety:** The model processes only placeholder tokens, preventing raw credential leakage into provider training datasets.

Stage 4: Validate (Critic & Bias Checker Agents + Controlled Retry)

- **Factual Critic Agent:** Dedicated LLM verifying claims against query context to eliminate hallucinations.
- **Bias & Fairness Checker:** Dedicated LLM auditing for stereotyping, unfair generalizations, and safety policy violations.
- **Governed Retry Loop:** If flagged, ControlPlane injects targeted feedback and re-prompts the agent up to MAX_RETRIES (default: 3).

Stage 5: Respond (Safe Detokenization & Delivery)

- **Collision-Free Detokenization:** Restores original plaintext PII strictly after validation passes.
- **Safe Final Output:** Delivers the verified response along with complete audit trail telemetry and latency metrics.

4. Economic Impact & Net Cost Optimization

By replacing blind frontier model routing with an optimized SLM agent paired with prompt compression and early risk gate short-circuiting:

Metric Dimension	Unconstrained Frontier LLM	ControlPlane.ai + SLM	Variance
Base Token Pricing	\$5.00 - \$15.00 / 1M tokens	\$0.18 / 1M tokens (Llama 3.1 8B)	-96.4%
Adversarial Query Cost	Full inference cost incurred	\$0.00 (Hard-blocked at Stage 1)	-100%
Prompt Token Consumption	100% (Uncompressed)	~65-75% (Fluff compressed)	-25-35%
Net Blended Cost	100% Baseline	47.1% Net Cost	~52.9% Savings

5. Execution Guide & Test Suite Verification

The project is fully tested (58 automated unit & end-to-end tests passing with 98% coverage) and demo-ready.

Launch Streamlit Dashboard: `.\controlplane\.venv\Scripts\streamlit.exe run controlplane\streamlit_app.py`

Run Automated Pytest Suite: `pytest tests/ -v --cov=controlplane`

GitHub Repository: <https://github.com/dhruvkachhela/controlplane>